

Media Information

8 Memorial Medical Court
Greenville, SC

PATHOLOGY
CONSULTANTS
specialized diagnostic services

Phone: (864) 295-3492
Fax: (864) 455-8926

IHC Prognostic Report

Patient
Weaver, Rachel

Gender: Female

MRN: 974694034

Ordering: Husman, MD, Abbie L.

03/24/1977

Collected: 09/02/2025

Received: 09/03/2025

Body Site: Right breast, 2:00, 3 cm from nipple, 6 cm mass, coil clip, ultrasound-guided core biopsy

C00499

WHC25-012873

Greenville Memorial

Phone: Fax: 864-455-8926

Treating: Shah, MD, Pavani Thotakura

Specimen: Formalin-fixed tissue

Clinical Information:

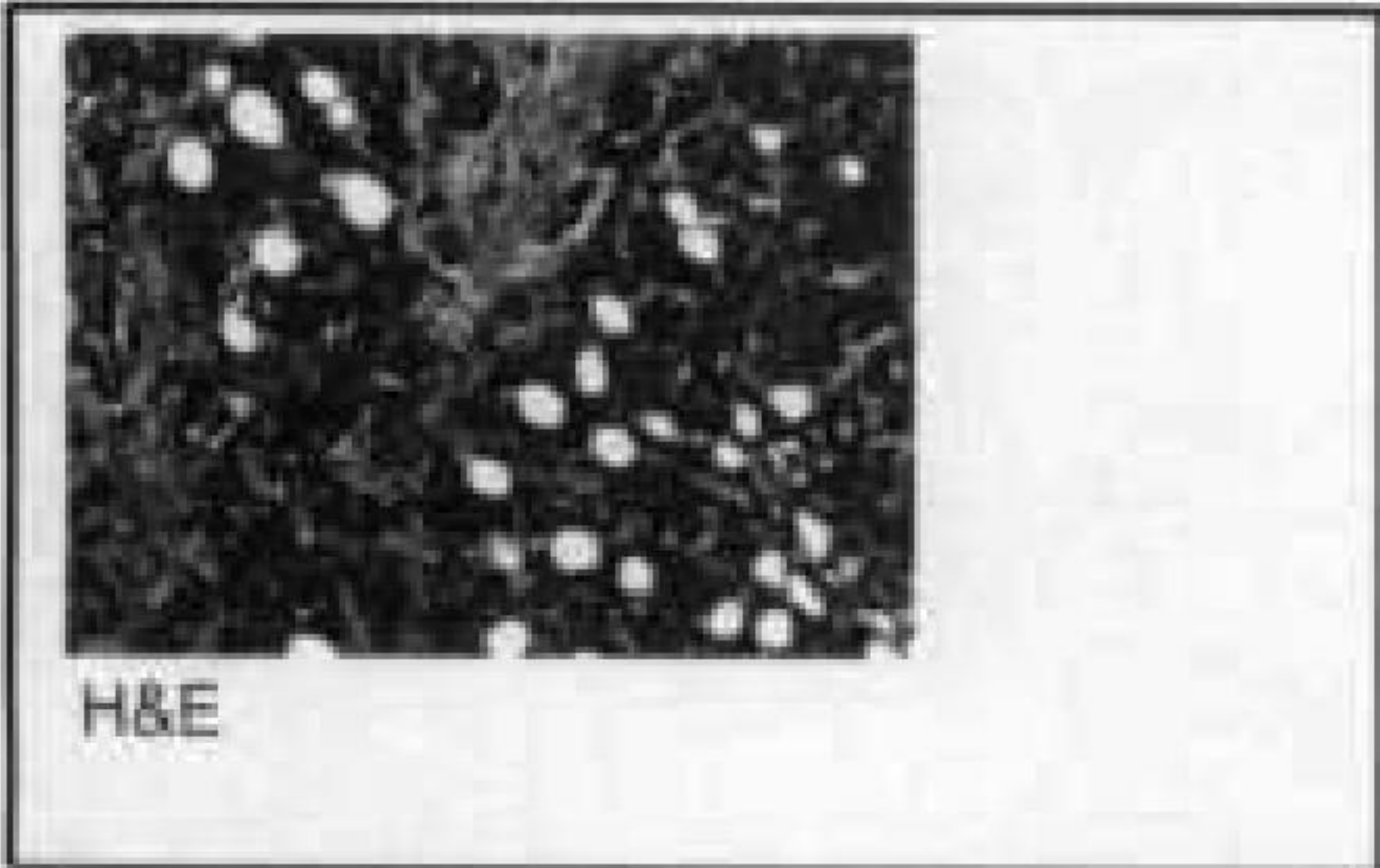
Malignant neoplasm of unspecified site of unspecified female breast; Unspecified lump in the right breast, unspecified quadrant; Invasive carcinoma of no

Cold Ischemia:

<1min

Fixation Time:

12-14hrs



 ER (SP1)	Overall Result:	Positive	ER (SP1) Reference Range	
	% Positive	86.7	< 1 %	Negative
	Staining Intensity	2	1 - 10 %	Low Positive
	Analysis	Image	> 10 %	Positive
	Assay Type	IVD		
	Clone	SP1		

 PR (1E2)	Overall Result:	Positive	PR (1E2) Reference Range	
	% Positive	85.4	< 1 %	Negative
	Staining Intensity	3	>= 1 %	Positive
	Analysis	Image		
	Assay Type	IVD		
	Clone	1E2		

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Patient Name: Patient ID: 746972 Accession:

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	Overall Result:	NEGATIVE (Ultra-Low)	HER2 (4B5) Reference Range	
	Score	0+	0	NEGATIVE (Null)
	Analysis	Image	0 +	NEGATIVE (Ultra-Low)
	Assay Type	LDT	1 +	NEGATIVE (Low)
	Clone	4B5	2 +	EQUIVOCAL
	Complete Intense Staining	0%	3 +	POSITIVE
	Her2 (4B5)			

	Overall Result:	High	Ki-67 Reference Range	
	% Positive	65.2	< 10 %	Low
	Staining Intensity	3	10 - 20 %	Intermediate
	Analysis	Image	%	
	Assay Type	IVD	> 20 %	High
	Clone	MIB-1		
	Ki-67 (MIB-1)			

Comments:

Clinical correlation is suggested.

Cold ischemia time and formalin fixation time meet ASCO/ CAP guidelines.

No present and appropriate.

IHC METHODOLOGY FOR BREAST CARCINOMA:
When performed, the following prognostic and/or predictive IVD markers were assessed by ready-to-use (RTU) primary monoclonal antibodies obtained from Ventana (Roche): (1) ER nuclear expression (clone SP1), (2) PR nuclear expression (clone 1E2), (3) p53 nuclear expression (clone Bp53-11), (4) HER2 membrane protein expression (clone 4B5), and (5) the Ki-67 nuclear proliferation marker employed a DAKO monoclonal antibody (clone MIB-1). These antibodies were stained on a Ventana automated platform utilizing an ultraView Universal Polymer Detection Kit. The percentage and intensity of ER/PR (0, 1+/weak, 2+/moderate, 3+/strong) of positively-stained tumor nuclei and the HER2 membrane staining score was determined either manually or in conjunction with a computer-assisted digital imaging system. In univariate analysis, moderate to strong staining of ER in even 1% of the invasive tumor cells is associated with significant improvement in disease-free survival compared to those patients whose tumors lack ER expression. In addition, although ER should be utilized as a predictor of benefit to adjuvant endocrine therapy, PR is used to only provide prognostic information to help stratify ER-positive patients whereby those with low/negative PR expressions have a worse prognosis. In terms of Ki67 interpretation, subcategories are defined by an international multicenter collaboration group. Unfavorable histopathologic features include: age < 50 years; negative ER and PR; ER (+) & PR low(-); a high proliferative index (Ki-67 > 20%); or angiolymphatic invasion, for example. In patients for whom anti-hormonal or HER2-targeted therapy is being considered, interpretative criteria are set forth by the latest ASCO/CAP Guidelines for breast cancer. In addition, the ASCO/CAP Guidelines also emphasize that the following parameters should be met: the tissue be histologically reviewed by a pathologist; all primary invasive, metastatic, and recurrent breast tumors be evaluated for HER2 status; validated ER, PR, and HER2 testing be performed on paraffin-embedded tissue optimally fixed in 10% NBF for 6-72 hours; the Cold Ischemic Time should be < 60 minutes; and unstained tissue sections should be ideally < 6 weeks old.

The percentage of tumor cells with uniform intense, complete membrane staining (2+ or 3+) is shown above. All controls are reviewed and show appropriate positive and negative immunoreactivity, unless otherwise specified. Since this IHC assay has not been validated on decalcified tissues, the results on decalcified specimens should be carefully interpreted given the possibility of false negativity.

The Reference ranges for breast carcinoma are as follows for the following markers (expressed as a percentage of cells with nuclear staining):

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Weaver, Rachel	03/24/1977	M25-35797-A1	C00499	WHC25-012873

BIOMARKER	RESULTS		
ER (SP1)	Negative (< 1%)	Low Positive (1% - 10%)	Positive (> 10% - 100%)
PR (1E2)	Negative (< 1%)		Positive (1% - 100%)
KI-67 (MIB-1)	Low Proliferation Rate (< 10%)	Intermediate Proliferation Rate (10% - 20%)	High Proliferation Rate (> 20% - 100%)
p53 (Bp53-11)	Negative (< 35%)		Positive (≥ 35% - 100%)

If the initial ER result is < 10% or the ER intensity is weak (1+), then internal control tissues are re-reviewed following ASCO/CAP 2020 Guidelines. The 'Low Positive' ER designation applies only to invasive carcinoma, and does not apply to PR expression or DCIS. When applicable, the following "Low Positive" ER comment is noted: the cancer in this sample has a low level (1-10%) of ER expression by IHC. There are limited data on the overall benefit of endocrine therapies for patients with low level (1-10%) ER expression, but they currently suggest possible benefit, so patients are considered eligible for endocrine treatment. There are data that suggest invasive cancers with these results are heterogeneous in both behavior and biology and often have gene expression profiles more similar to ER negative cancers.

Based on the latest ASCO/CAP 2020 ER/PR Guidelines, ER testing should also be performed on DCIS tumors without invasion since there is growing clinical literature that adjuvant endocrine therapy can reduce the risk of future breast cancer in these patients. Although only ER should be used to select these patients for endocrine therapy, reporting PR results may still provide prognostic information and is considered optional in DCIS patients.

With selected invasive breast cancer subtypes, ER results are usually histologically concordant with these tumors. Thus, when tumor histology is concordant with ER results (> 10% staining with 2+/3+ intensity), the result may be reported as 'ER-POSITIVE'. But if the tumor histology is concordant with the ER results AND shows 'Weak Staining Intensity (1+)' OR only 0-10% cells stain with ER, then the results must be adjudicated, per Guidelines. Lastly, if discordant (or highly unusual) ER results are obtained in relation to the tumor histologic subtype, then a second pathologist review (or digital analysis) and repeat testing has been performed, if necessary. These tumors include: (1) 'Discordant tumors that are aberrantly ER-POSITIVE' (for example, Metaplastic carcinomas; Adenoid cystic carcinoma including other salivary-like carcinomas of the breast; Secretory carcinoma; or Carcinomas with Apocrine differentiation), and (2) 'Discordant tumors that are aberrantly ER-NEGATIVE' (for example, LG invasive ductal carcinomas of no special type; Lobular carcinoma, classic type; Pure tubular, cribriform or mucinous carcinomas; Encapsulated papillary carcinoma; or Solid Papillary carcinomas).

In a recent study (Mueßer, 2023), an optimal p53 index cutoff of ≥ 35% was utilized to help identify luminal breast cancer with TP53 mutations in treatment-naïve patient specimens (with high specificity of 92% and accuracy of 87%). In contrast, in endocrine pre-treated patients, the cutoff was 25% (with high specificity of 95% and accuracy of 90%).

Scoring Criteria of HER2 (4B5) results for invasive breast carcinoma are subclassified as follows:

HER2 IHC Score	Staining Pattern of HER2 IHC results for Breast Carcinoma	HER2 IHC Overall Result
0 (Without Staining)	No membrane staining is observed (Null)	Negative (Null)
0+ (With Staining)	Membrane staining that is incomplete and is faint/barely perceptible in ≤ 10% of invasive tumor cells.	Negative (Ultra-Low)
1+	Incomplete membrane staining that is faint/barely perceptible in > 10% of invasive tumor cells	Negative (Low)*
2+	Weak to moderate complete membrane staining in > 10% of tumor cells OR Complete membrane staining that is intense but ≤ 10% of tumor cells.	Equivocal
3+	Circumferential membrane staining that is complete, intense, and > 10% of invasive tumor cells	Positive

*HER2-Low encompasses both HER2 IHC 1+ and IHC 2+/FISH non-amplified.

In accordance with the CAP Biomarker Reporting protocol, HER2 tests results of invasive breast cancer samples with either 2+ (Equivocal) or 3+ (Positive) should also provide the percentage of tumor cells with uniform, intense, complete membrane staining (3+ percentage). In addition, equivocal (2+) IHC findings will be automatically reflexed to 'HER2/CEP17' FISH testing'. The HER2 FISH-criteria remain unchanged as described in the ASCO/CAP Guidelines (2023), and the CAP Cancer Protocol for Breast Biomarker Testing (2025).

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Patient Name: [REDACTED]

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When the initial invasive needle core biopsy is HER2 Negative (IHC or FISH) or the biopsy is not representative of the entire mass, then a new HER2 test may be ordered if one of the following is observed: (1) Tumor is grade 3; (2) Amount of invasive tumor in the initial core biopsy specimen is small; (3) Resection specimen contains high-grade carcinoma that is morphologically distinct from that in the core; (4) The core biopsy result is equivocal (2+) for HER2 by both IHC and FISH; (5) There is doubt about the handling of the core biopsy specimen (long ischemic time, short time in fixative, different fixatives); or (6) the test is considered "Indeterminate". A specimen is deemed "Indeterminate" if any of these limitations are met: (a) fixative and/or suboptimal fixation times that are outside of these established pre-analytic guidelines (fixation aberrancy); (b) processing, crush or edge artifacts, when present; (c) external positive control tissues do not stain as expected; (d) internal control tissue (such as, normal breast ductules) shows aberrant strong membrane or cytoplasmic reactivity (except for positive benign apocrine metaplasia); (e) non-formalin fixatives utilized (unless previously validated); (f) decalcified specimens without any immunoreactivity; or (g) insufficient tumor. Thus, these results should be reported either as: "Indeterminate" with no [REDACTED] provided as shown above, they should be interpreted with caution due to possible preservation issues.

HER2-LOW AND ULTRA-LOW BREAST CANCER: Based on recent clinical trials (phase 3 DESTINY-Breast06 trial, NCT04494425), the use of "Enhertu" (fam-trastuzumab deruxtecan-nxki; T-DXd), may be utilized for several indications. These include the treatment of adult patients with unresectable or metastatic HR+/HER2-LOW (HER2 IHC 1+ or IHC 2+/FISH not amplified), HER2-ULTRA-LOW (IHC 0+ with membrane staining identified) breast cancer that has progressed on one or more endocrine therapies in the metastatic setting, or HER2-positive breast cancer (3+, or ISH positive). Enhertu is also currently approved for use in gastric, GEJ, solid tumors, and lung carcinomas with specific indications.

Patients with breast cancers that are HER2 IHC 3+ or ISH amplified may be eligible for several therapies that disrupt HER2 signaling pathways.

HER2 FISH CRITERIA:

DUAL PROBE FISH ASSAY	Group 1	Group 2	Group 3	Group 4	Group 5
HER2:CEP 17 ratio	≥ 2.0 AND	≥ 2.0 AND	< 2.0 AND	< 2.0 AND	< 2.0 AND
Average HER2 signals/cell	≥ 4.0 signals/cell	< 4.0 signals/cell	≥ 6.0 signals/cell	≥ 4.0 and < 6.0 signals/cell	< 4.0 signals/cell

HER2 FISH GROUPINGS:

HER2 POSITIVE Categories	HER2 NEGATIVE Categories
GROUP 1 (HER2-Positive regardless of IHC result)	
GROUP 2 AND Concurrent IHC 3+	GROUP 2 AND Concurrent IHC 0, 1+ or 2+
GROUP 3 AND Concurrent IHC 2+ or 3+	GROUP 3 AND Concurrent IHC 0, 1+
GROUP 4 AND Concurrent IHC 3+	GROUP 4 AND Concurrent IHC 0, 1+ or 2+
	GROUP 5
IHC ≥ 3+ (HER2-Positive regardless of FISH result)	

References:

1. Bardia A. N Engl J Med. 2024;391(22):2110-2122.
2. Wolff AC. Arch Pathol Lab Med. 2020;144(5):593-1000.
3. Leung SCY. NPJ Breast Cancer. 2018 May 18;2:16014.
4. Allison KH. Arch Pathol Lab Med. 2020 May;144(5):545-563.
5. Allison K. CAP Cancer Protocols. 2025 March; Version 1.6.0.0.
6. Enhertu. Prescribing information. Daiichi Sankyo Inc. 2025.
7. Franchina M. Int J Mol Sci. 2023;24(16):12796.
8. Mueller S. Mod Pathol. 2023;36(4):[REDACTED]

Electronically Signed by: Kari Valente M.D. on

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Patient [REDACTED]

Document Information

Other: Outside Pathology

PATH

██████████ 00:00

Attached To:

OUTSIDE PATHOLOGY ORDER ██████████

Orders Only on ██████████ with Historical Provider, ██████████

Source Information

Jaslee Quinones | Msk Pas